

2. ENVIRONMENTAL MANAGEMENT

Evolution of Indian environmental policy

The concept of environmental protection is an age old idea imbibed in the Indian cultural ethos since time immemorial.

But during the Pre-Independence Era (Before 1947), environmental concerns were not a primary focus during this period

To understand the present-day legal system for environment protection and conservation of natural resources, it is important to look into the past Indian traditions and practices of protecting the environment. The India Constitution adopted in 1950 did not deal with the subject of environment or prevention and control of pollution as such. It was the Stockholm Declaration of 1972 which turned the attention of the Indian Government to the broader perspective of environmental protection.

India's environmental policy took shape in the 1970s and 1980s when environmental obligations were included in the constitution and federal laws were framed.

Environmental policy is defined as “any action deliberately taken to manage human activities with a view to prevent, reduce, or mitigate harmful effects on nature and natural resources, and ensuring that man-made changes to the environment do not have harmful effects on humans or the environment,” (McCormick 2001)

It is the commitment of an organization to the laws, regulations, and other policy mechanisms concerning environmental issues. The ISO 14001 standard states that an environmental policy is the organisation's overall environmental performance intentions and direction formally expressed by top management.

The environmental issues generally air , soil and water pollution, waste management, ecosystem management, maintenance of biodiversity, the protection of natural resources, wildlife and endangered species.

Benefits of developing an environmental policy –

- ☐ An organisation can publicly advertise its environmental performance and adopted best practice or is working towards improving its environmental performance
- ☐ It can provide clear direction to all stakeholders about the organisation's environmental status and environmental objectives

Development of an environmental policy –

Managing team along with some experts in this field can develop the EMS of the organization.

Environmental Law is a complex combination of state, federal, and international treaty law pertaining to issues of concern to the environment and protecting natural resources.

These laws actually regulate the impact of human activities on the environment. Environmental law covers a broad range of activities that affect air, water, land, flora or fauna.

Environmental legislation –

It is a collection of many laws and regulations aimed at protecting the environment from harmful actions.

Need for environmental legislation –

- To safeguard our environment from further degradation and pollution
- To curb the destructive practices of unscrupulous people, forest mafia groups, poachers, polluters and over exploitation of environmental resources

Constitutional Provisions

By the Forty Second Amendment Act of 1976, Article 51A(g) was added to Indian constitution. Article 51-A (g) which deals with Fundamental Duties of the citizens states: “It shall be the duty of every citizen of India to protect and improve the natural environment including forests, lakes, rivers and wildlife and to have compassion for living creatures.”

Directive Principles of State Policy regarding the protection and improvement of the environment. Article 48A states “The State shall endeavour to protect and improve the environment and to safeguard the forests and wildlife of the country”.

During the United Nations Conference on Human Environment at Stockholm, 1972 Mrs. Indira Gandhi, the then Prime Minister of India declared that “The natural resources of the earth including the air ,water, land flora and fauna must be safeguard for the benefit of present and future generations through careful planning and management. Nature conservation must therefore receive importance in planning for economic development”

UN Conferences and commissions

➤ **United Nations Conference on Human Environment (UNCHE)**

The participants adopted a series of principles for sound management of the environment including the Stockholm Declaration and Action Plan for the Human Environment and several resolutions. The Action Plan contained three main categories: a) Global Environmental Assessment Programme (watch plan); b) Environmental management activities; (c) International measures to support assessment and management activities carried out at the national and international levels.

- ❑ held at Stockholm during June 5-14, 1972
- ❑ also known as Stockholm Conference
- ❑ the first major international conference that focused on international environmental issues
- ❑ led to the creation of the United Nations Environment Programme (UNEP) in December 1972 to coordinate global efforts to promote sustainability and safeguard the natural environment
- ❑ declared 5th June as World Environment Day
- ❑ Stockholm Declaration containing 26 principles on environment and development was adopted

➤ World Commission on Environment and Development (WCED)

- ❑ Also known as Brundtland Commission
- ❑ created by United Nations in 1983
- ❑ aimed to create a united international community with shared sustainability goals by identifying sustainability problems worldwide, raising awareness about them, and suggesting the implementation of solutions
- ❑ *Our Common Future*, also known as the *Brundtland Report*, released in 1987 popularized and defined sustainable development



➤ **United Nations Conference on Environment and Development (UNCED)**

- ❑ The Earth Summit, held during June 3- 14, 1992 at Rio de Janeiro, Brazil
- ❑ principal theme was environment and sustainable development
- ❑ Agenda 21, the Rio Declaration on Environment and Development, the Statement of Forest Principles, the United Nations Framework Convention on Climate Change and the United Nations Convention on Biological Diversity are the outcomes.
- ❑ The Rio de Janeiro conference brought to light the interdependence and progressive evolution of many social, economic, and environmental elements, as well as the necessity of action in other areas to sustain success in one area over an extended period of time.
- ❑ The primary objective of the Rio 'Earth Summit' was to produce a broad agenda and a new blueprint for international action on environmental and development issues that would help guide international cooperation and development policy in the twenty-first century



➤ World Summit on Sustainable Development (WSSD)

- ❑ Also known as Rio +10 was held during August 26 to September 4, 2002
- ❑ The 2002 World Summit on Sustainable Development Johannesburg adopted a Political Declaration and Implementation Plan which included provisions covering a set of activities and measures to be taken in order to achieve development that takes into account respect for the environment
- ❑ held at Johannesburg, South Africa.
- ❑ This Summit saw the participation of more than a hundred heads of state and government and tens of thousands of government representatives and non-governmental organizations, resulted, after several days of deliberations, decisions that related to water, energy, health, agriculture, biological diversity and other areas of concern.
- ❑ an unprecedented opportunity to strengthen the role of health in sustainable development.



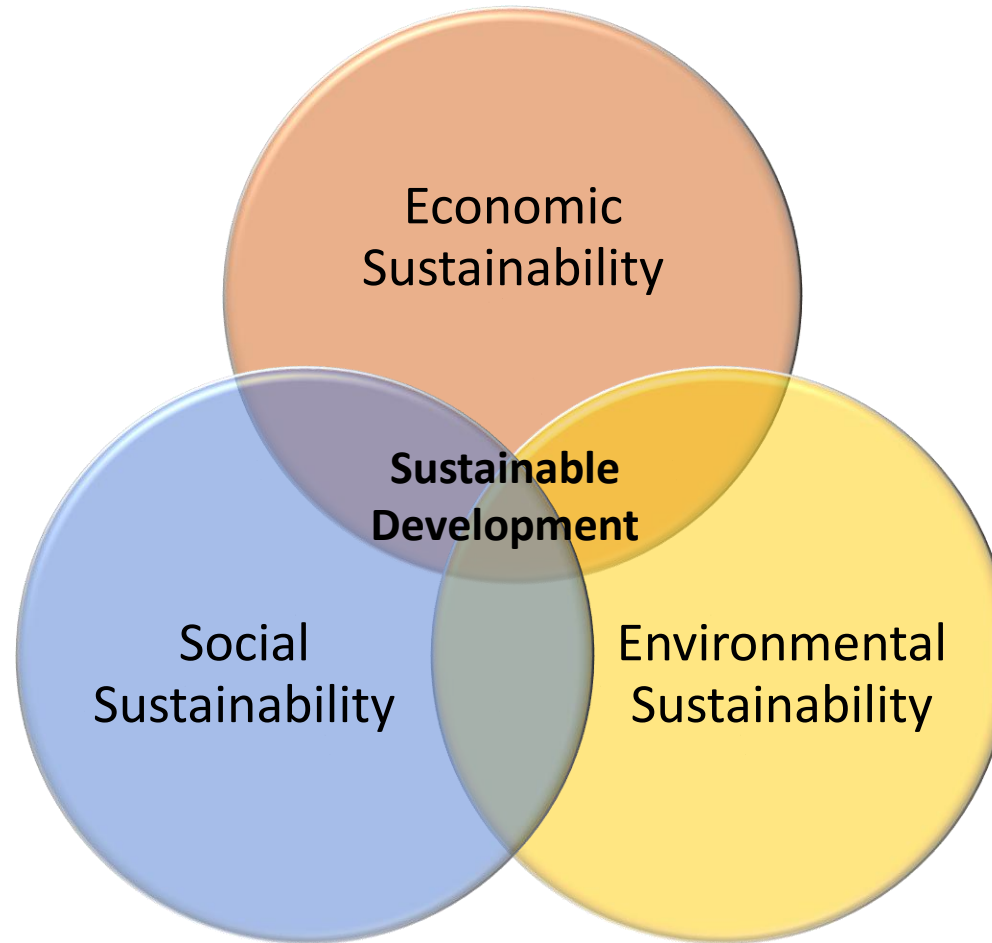
The concept of sustainable development has been recognized in 1987 when the World Commission on Environment and Development (The Brundtland Commission) introduced the term Sustainable Development in its report Our Common Future for the development that meets the needs of the present without compromising on the ability of future generations to meet their own needs.

Agenda 21, adopted during the United Nations Conference on Environment and Development (Earth Summit) held in Rio de Janeiro, Brazil in 1992 is a blueprint for achieving sustainable development.

Components of sustainable development –

- ☐ Economic sustainability
- ☐ Social sustainability
- ☐ Ecological sustainability





economic and social development which are conducted without causing environmental degradation.

United Nations Conference on Sustainable Development, 20-22 June 2012, Rio de Janeiro

- ❑ The 2012 United Nations Conference on Sustainable Development in Rio (also known as Rio+20) resulted in a document containing clear and practical steps for the implementation of sustainable development.
- ❑ At the Conference, Member States decided to launch a process to develop a set of Sustainable Development Goals (SDGs), building on the Millennium Development Goals (MDGs) and converging with the post-2015 development agenda.

The Conference also adopted innovative guidelines on green economy policies, and put in place a strategy for financing sustainable development.



Convention on Long-range Transboundary Air Pollution (CLRTAP)



Aims

- to limit and gradually reduce and prevent air pollution including long-range transboundary air pollution
- to combat the discharge of air pollutants through exchanges of information, consultation, research and monitoring.

The convention opened for signature on November 13, 1979, and entered into force on March 16, 1983. LRTAP is a convention in the United Nations Economic Commission for Europe (UNECE). As the first regional environmental convention, it has been instrumental in the reduction of key harmful pollutants in both Europe and North America. Over the past 30 years, CLRTAP has been extended by 8 Protocols aimed at setting strict reduction targets for releases of pollutants for the protection of human and environmental health and each of these Protocols targets pollutants such as sulphur, nitrogen oxide, persistent organic pollutants, volatile organic compounds, ammonia, and toxic heavy metals.

Convention on Biological Diversity



The Convention on Biological Diversity (CBD) is an international legally-binding treaty. The Convention was opened for signature at the Earth Summit in Rio de Janeiro on 5 June 1992 and entered into force on 29 December 1993. This Convention is now one of the most widely ratified international treaties on environmental issues, with 194 member countries. At the 2010 10th Conference of Parties (COP) to the Convention on Biological Diversity in October in Nagoya, Japan, the Nagoya Protocol was adopted. It mainly aims at developing national strategies for the conservation and sustainable use of biological diversity.

Objectives:

- conservation of biological diversity,
- sustainable use of its components,
- fair and equitable sharing of the benefits arising out of the utilisation of genetic resources

One of the **CBD's greatest achievements** so far has been to generate an enormous amount of interest in biodiversity, both in developed and developing countries. Biodiversity is now seen as a critically important environment and development issue.

Nagoya Protocol

The Nagoya Protocol on Access to Genetic Resources and the Fair and Equitable Sharing of Benefits Arising from their Utilization (ABS) to the Convention on Biological Diversity is a supplementary agreement to the Convention on Biological Diversity.

The Nagoya Protocol applies to genetic resources that are covered by the CBD, and to the benefits arising from their utilization. The Nagoya Protocol also covers traditional knowledge (TK) associated with genetic resources that are covered by the CBD and the benefits arising from its utilization.



Objective of this protocol is the fair and equitable sharing of the benefits arising from the utilization of genetic resources, including by appropriate access to genetic resources and by appropriate transfer of relevant technologies, taking into account all rights over those resources and to technologies, and by appropriate funding, thereby contributing to the conservation of biological diversity and the sustainable use of its components.

Importance of Nagoya Protocol :

It will create greater legal certainty and transparency for both providers and users of genetic resources by:

- Establishing more predictable conditions for access to genetic resources.
- Helping to ensure benefit-sharing when genetic resources leave the country providing the genetic resources

The Cartagena Protocol on Biosafety



- The Cartagena Protocol on Biosafety to the Convention on Biological Diversity was adopted on 29 January 2000 and entered into force on 11 September 2003.
- It is a supplementary agreement to the United Nations Convention on Biological Diversity (CBD). It is one of the important international environmental protocols on biodiversity
- Objective: to contribute to ensuring an adequate level of protection in the field of the safe transfer, handling and use of living modified organisms resulting from modern biotechnology that may have adverse effects on the conservation and sustainable use of biological diversity, taking also into account risks to human health, and specifically focusing on transboundary movements
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- It provides a comprehensive and holistic approach to the conservation of biological diversity, the sustainable use of natural resources and the fair and equitable sharing of benefits deriving from the use of genetic resources



CITES (Convention on International Trade in Endangered Species of Wild Fauna and Flora) is a global agreement among governments to regulate or ban international trade in species under threat. It is a multilateral treaty to protect endangered plants and animals from the threats of international trade.

- Concerned about the overexploitation of many vulnerable species as a result of unregulated international trade, governments adopted the Convention on International Trade in Endangered Species of Wild Fauna and Flora (CITES) in 1973 and the treaty entered into force in 1975.
- CITES regulates international trade in over 36,000 species of plants and animals, including their products and derivatives, ensuring their survival in the wild with benefits for the livelihoods of local people and the global environment.

TRIPS Agreement

The Agreement on Trade-Related Aspects of Intellectual Property Rights (TRIPS) is an international legal agreement between all the member nations of the World Trade Organization (WTO).

Adopted in April, 1994 and entered into force in January 1995

The TRIPS agreement introduced intellectual property law into the multilateral trading system for the first time and remains the most comprehensive multilateral agreement on intellectual property to date.

TRIPS establishes minimum standards for the availability, scope, and use of seven forms of intellectual property: copyrights, trademarks, geographical indications, industrial designs, patents, layout designs for integrated circuits, and undisclosed information (trade secrets)

Vienna Convention for the Protection of the Ozone Layer

It is a framework agreement in which States agree to cooperate in relevant research and scientific assessments of the ozone problem, to exchange information, and to adopt “appropriate measures” to prevent activities that harm the ozone layer. The obligations are general and contain no specific limits on chemicals that deplete the ozone layer.

It was adopted on 22 March 1985 and entered into force on 22 September 1988

The Vienna Convention is an important part of the international ozone regime, providing the forum for discussions on scientific research and observations of the ozone layer.

This convention aims to promote cooperation among nations by exchanging information on the effects of human activities on the ozone layer.

Vienna Convention requires parties to take appropriate measures which include

- adoption of legislative and administrative measures;
- exchange of information;
- cooperation on research and scientific assessment;
- development and transfer of technology.



Carbon trading – a market based tool to limit GHGs

Carbon trading is an approach used to control carbon dioxide (CO₂) pollution by providing economic incentives for achieving emissions reductions.

It is sometimes called **cap and trade** or **carbon emissions** trading. Under a basic cap-and-trade scheme, if a company's carbon emissions fall below a set allowance, that company can sell the difference — in the form of credits — to other companies that exceed their limits.

Another fast-growing voluntary model is **carbon offsets**. In this global market, a set of middlemen companies, called offset firms, estimate a company's emissions and then act as brokers by offering opportunities to invest in carbon-reducing projects around the world. Unlike carbon trading, offsetting isn't yet government regulated in most countries.

The European Union's Emissions Trading System (ETS) is the world's biggest scheme for trading greenhouse gas emissions allowances.

Carbon credit - a permit which allows a country or organization to produce a certain amount of carbon emissions and which can be traded if the full allowance is not used.

It is a generic term for any tradable certificate or permit representing the right to emit one tonne of carbon dioxide or the mass of another greenhouse gas with a carbon dioxide equivalent (tCO₂e) equivalent to one tonne of carbon dioxide.

International Organizations



The United Nations Environment Programme (UNEP) is the leading global environmental authority that sets the global environmental agenda, promotes the coherent implementation of the environmental dimension of sustainable development within the United Nations system and serves as an authoritative advocate for the global environment.

- ❑ strengthens the ability of countries to integrate climate change responses by providing leadership in adaptation, mitigation, technology and finance
- ❑ conducts environmental assessments in crisis-affected countries and provides guidance for implementing legislative and institutional frameworks for improved environmental management.
- ❑ strives to minimize the impact of harmful substances and hazardous waste on the environment and human beings
- ❑ focuses on resource efficiency/ sustainable consumption and production
- ❑ facilitates management and restoration of ecosystems in a manner consistent with sustainable development, and promotes use of ecosystem services.
- ❑ supports governments in establishing, implementing and strengthening the necessary processes, institutions, laws, policies and programs to achieve sustainable development at the country, regional and global levels, and mainstreaming environment in development planning



It was founded in 1948. WHO (World Health Organisation) is the directing and coordinating authority for health within the United Nations system.

Main functions of WHO:

- ☐ providing leadership on matters critical to health and engaging in partnerships where joint action is needed;
- ☐ shaping the research agenda and stimulating the generation, translation and dissemination of valuable knowledge;
- ☐ setting norms and standards and promoting and monitoring their implementation;
- ☐ articulating ethical and evidence-based policy options; providing technical support, catalysing change, and building sustainable institutional capacity;
- ☐ monitoring the health situation and assessing health trends



UNDP, headquartered in New York City, works in almost 170 countries and territories. It has worked in India since 1951 in almost all areas of human development

- Focuses on sustainable development, democratic governance and peace-building, climate and disaster resilience
- Works for eradication of extreme poverty, and reducing inequalities and exclusion
- supports national processes to accelerate the progress of sustainable human development
- protection of human rights and the empowerment of women in all of its programmes
- assisting developing countries in implementing environmentally sound policies and practices.

Food and Agriculture Organization (FAO)



Food and Agriculture Organization is a specialized organization of the United Nations that leads to international efforts to defeat hunger.

- ☐ Help eliminate hunger, food insecurity and malnutrition
- ☐ Make agriculture more productive and sustainable
- ☐ Ensure inclusive and efficient agricultural and food systems
- ☐ Protect livelihoods from disasters
- ☐ Reduce rural poverty and improve rural livelihood



International Union for Conservation of Nature (IUCN) is an international organization working in the field of nature conservation and sustainable use of natural resources.

Functions:

- To tackle three of the most important challenges facing the world's ocean and polar regions: climate change, biodiversity loss and pollution.
- To support countries and communities in achieving effective and equitable protected and conserved areas that yield positive outcomes for society.
- to develop best practices and approaches that enable effective conservation and help sites achieve high standards, while also informing professional capacity development and influencing national and global policy.
- To evaluate sites nominated for the World Heritage List and monitors the state of conservation of listed sites, aiming to improve the management of World Heritage sites and enhance the role of the Convention in nature conservation and sustainable development

The Ministry of Environment, Forest and Climate Change (MoEFCC)

It is the nodal agency in the administrative structure of the Central Government for the planning, promotion, co-ordination and overseeing the implementation of India's environmental and forestry policies and programmes.



The broad objectives of the Ministry are :

- Conservation and survey of flora, fauna, forests and wildlife
- Prevention and control of pollution
- Afforestation and regeneration of degraded areas
- Protection of the environment and
- Ensuring the welfare of animals

MoEFCC is primarily concerned with implementation of policies and programmes relating to conservation of the country's natural resources including its lakes and rivers, its biodiversity, forests and wildlife, ensuring the welfare of animals, and the prevention and abatement of pollution.



- ☐ Provide advice to the Central Government on any issue pertaining to air and water pollution mitigation and control, as well as air quality improvement
- ☐ Create and oversee the implementation of a national programme aimed at preventing, controlling, or reducing air and water pollution
- ☐ Coordinate State Board operations and settle disagreements among members
- ☐ Collect, compile and publish technical and statistical data relating to water and air pollution and the measures devised for their effective prevention, control or abatement
- ☐ Organize comprehensive mass awareness programme on the prevention, control or abatement of water and air pollution through mass media and disseminate information on matters relating to water and air pollution and their prevention and control

West Bengal Pollution Control Board (WBPCB)

The WBPCB is the frontal organization of the Environment Department for regulating with environmental pollution problems and their control in the State



- ☐ to carry out such investigations as may be directed by the Central Government to lay down standards for the quality of environment and discharge of environmental pollutants, to monitor and to enforce the standards laid down.
- ☐ to co-ordination between departments and agencies of the State and Union Governments concerned with policies and schemes relating to environment
- ☐ to send reports regarding its activities to the Central Government.
- ☐ to advise the State Government on any matter concerning the prevention, control or abatement of water pollution

The Water (Prevention and Control of Pollution) Act of 1974

The Water (Prevention and Control of Pollution) Act was enacted in 1974 to provide for the prevention and control of water pollution, and for the maintaining or restoring of wholesomeness of water in the country. The Act was amended in 1988.

Objectives:

- ☐ Prevent and control water pollution.
- ☐ Maintain and restore the wholesomeness of water.
- ☐ Establish central and state authorities for monitoring and regulation.
- ☐ Coordinates efforts across states for shares water resources

As per this Act, the Central Pollution control Board advises the Central Government on any matter concerning the prevention and control of water pollution, coordinate the activities of the State Boards and resolve disputes among them and organize comprehensive program regarding the prevention and control of water pollution through mass media.

State Pollution Control Boards are supposed to plan a comprehensive program for preventing and controlling the pollution of the wells and streams in the state and to secure its execution and advise the State Government on matters relating to prevention and controlling water pollution. They collaborate with the

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The Air (Prevention and Control of Pollution) Act of 1981

This Act aims to improve the quality of air and to prevent, control and abate air pollution in the country

An air pollutant is any solid, liquid or gaseous substance present in the atmosphere in such concentration as may be or tend to be injurious to human beings or other living creatures or plants or property or environment.

The Act enable an integrated approach to environmental problems, the Air Act expanded the authority of the central and state boards established under the Water Act, to include air pollution control. This Act prohibits the discharge of PM, lead, nitrogen oxide, sulphur oxide VOC, carbon monoxide and other pollutants into the air beyond a certain limit.

The Act requires an approval to any industrial plant prior to its operation and the SPCBs are assigned with the responsibility for monitoring the pollution levels.

As per this Act, the main functions of the State Pollution Control Boards are to plan a comprehensive programme for prevention, control and abatement of air pollution and to secure the execution thereof. To advise the State Government on any matter concerning prevention, control and abatement of air pollution.

The Environment (Protection) Act of 1986

The Act is an “umbrella” legislation that has provided a framework for the environmental regulation regime in India, which covers all major industrial and infrastructure activities and prohibits and regulates specific activities in coastal areas and eco-sensitive areas. It establishes the framework for studying, planning, and implementing long-term requirements of environmental safety and laying down a system of speedy and adequate response to situations threatening the environment.

Background:

The Environment (Protection) Act, 1986 finds its origin in the United Nations Conference on the Human Environment held in Stockholm in June 1972, where India was a participant. The Act was introduced to put into effect the commitments and resolutions adopted during that conference, aimed at improving the quality of the human environment.

Aims:

- to regulate industrial locations,
- to prescribe procedures for managing hazardous substances;
- to establish safeguards preventing accidents, and to collect and dismantle information regarding environmental pollution

This law extends to the whole of India.

The Environment (Protection) Act of 1986

The law gives power to central government to undertake any measures or steps which are necessary for protecting and improving the environment.

- By virtue of this Act, Central Government has armed itself with considerable powers which include, coordination of action by state, planning and execution of nation wide programmes, laying down environmental quality standards, specially those governing emission or discharge of environmental pollutants, placing restriction on the location of industries and so on.
- This Act makes EIA mandatory for the specified 29 industries.
- This law prevents the reckless handling of hazardous substances except as is prescribed or stated.
- It also provides provision for penalties

Penalties:

Imprisonment up to 5 years or fines up to ₹1 lakh for violations.
Additional fines for continuing offenses

Significance

This act enabled creation of rules like:

- Hazardous Waste Management Rules
- Biomedical Waste Rules
- Coastal Regulation Zone Rules
- Strengthened India's ability to respond to environmental crises and regulate industries.

Wildlife Protection Act, 1972

The Act provides for the protection of wild animals, birds and plants; and for matters connected therewith or ancillary or incidental thereto. It extends to the whole of the country, except the State of Jammu and Kashmir which has its own wildlife act.

Objectives:

Provide a legal framework for the protection of India's biodiversity.

- ☐ Prevent hunting and poaching of wild animals.
- ☐ Regulate trade in wildlife and animal articles.
- ☐ Establish protected areas like National Parks, Wildlife Sanctuaries, and Conservation Reserves.
- ☐ Align India's conservation efforts with global conventions like CITES (Convention on International Trade in Endangered Species)

Significance

- First comprehensive legislation for wildlife conservation in India.
- Basis for creation of hundreds of National Parks and Sanctuaries.
- Strengthened India's role in global biodiversity conservation.
- Continues to evolve through amendments (e.g., 2002, 2006, 2022) to address modern challenges.

Forest policy and Forest Conservation Act, 1980

This act is a unique piece of legislation and regulatory mechanism that reflects the collective will of the nation to protect its rich forests, biodiversity and natural heritage and resources. The act permits only unavoidable use of forest land for various development purposes. It embodies the firm commitment of the Government of India to balance the conservation of forests with the sustainable development need of the country contributing to better environment, health and economy.

This act aims to provide for the conservation of forest and for matters connected therewith or ancillary or incidental thereto

The 1989 amendment permitted the forest officers to be prosecuted and to be subjected to fine, similar to other persons.

The main components of social forestry are :

- ❖ community woodlots on common village lands
- ❖ strip plantations, along the roadsides and canal banks
- ❖ farm forestry



National Green Tribunal (NGT)

The NGT was established on October 18, 2010 under the National Green Tribunal Act 2010, passed by the Central Government.

The legislature Act of Parliament defines the National Green Tribunal Act, 2010 as follows,

"An Act to provide for the establishment of a National Green Tribunal for the effective and expeditious disposal of cases relating to environmental protection and conservation of forests and other natural resources including enforcement of any legal right relating to environment and giving relief and compensation for damages to persons and property and for matters connected therewith or incidental thereto"

Objectives:

to provide a specialized forum for effective and speedy disposal of cases pertaining to environment protection, conservation of forests and for seeking compensation for damages caused to people or property due to violation of environmental laws or conditions specified while granting permissions.

Environmental movements

- Also known as green movement or conservation movement.
- a social or political movement, for the conservation of environment or for the improvement of the state of the environment
- favor the sustainable management of natural resources and stress the protection of the environment via changes in public policy
- range from the highly organized and formally institutionalized ones to the radically informal activities.
- generally centred on ecology, health and human rights
- environmental movements range from being local to the almost global

Major reasons of the emergence of environmental movements in India include:

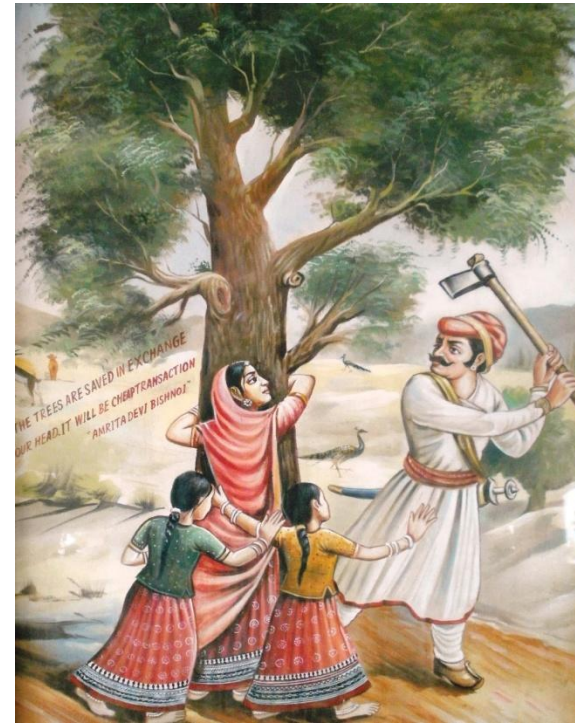
- ☐ Control over natural resources
- ☐ Socio –economic reasons and false developmental policies of the government
- ☐ Environmental degradation/ destruction
- ☐ Spread of environmental awareness

Bishnoi movement

Place: Khejarli, Marwar region,
Rajasthan state.

Leaders: Amrita Devi along with Bishnoi
villagers in Khejarli and surrounding
villages.

Aim: Save sacred trees from being cut
down by the king's soldiers for a new
palace.



Amrita Devi, a female villager could not bear to witness the destruction of both her faith and the village's sacred trees. She hugged the trees and encouraged others to do the same. 363 Bishnoi villagers were killed in this movement. The Bishnoi tree martyrs were influenced by the teachings of Guru Maharaj Jambaji, who founded the Bishnoi faith in 1485 and set forth principles forbidding harm to trees and animals. The king who came to know about these events rushed to the village and apologized, ordering the soldiers to cease logging operations. Soon afterward, the maharajah designated the Bishnoi state as a protected area, forbidding harm to trees and animals. This legislation still exists today in the region.

Chipko movement

- ❑ Place: In Chamoli district and later at Tehri-Garhwal district of Uttarakhand.
- ❑ Leaders: Sundarlal Bahuguna, Gaura Devi, Sudesha Devi, Bachni Devi, Chandi Prasad Bhatt, Govind Singh Rawat, Dhoom Singh Negi, Shamsheer Singh Bisht and Ghanasyam Raturi.
- ❑ Aim: The main objective was to protect the trees on the Himalayan slopes from the axes of contractors of the forest.



The Chipko movement is famous for its tree-hugging campaign to resist the tree cutting. It was started by noted environmentlist Sunderlal Bahuguna in 1970 to safe guard the rich forest of western Himalaya Range. He enlightened the villagers by conveying the importance of trees in the environment which checks the erosion of soil, cause rains and provides pure air. The women of Advani village of Tehri-Garhwal tied the sacred thread around trunks of trees and they hugged the trees, hence it was called '**Chipko Movement**' or 'hug the tree movement'. The main demand of the people in these protests was that the benefits of the forests (especially the right to fodder) should go to local people. The Chipko movement gathered momentum in 1978 when the women faced police firings and other tortures. State government took initiative to set up a committee to look into the matter, which eventually ruled in favor of the villagers. This became a turning point in the history of eco-development struggles in the region and around the world.

Silent Valley Movement

Place: Silent Valley, an evergreen tropical forest in the Palakkad district of Kerala, India.

Leaders: The Kerala Sastra Sahitya Parishad (KSSP) an NGO, and the poet-activist Sughathakumari played an important role in the Silent Valley protests.

Aim: In order to protect the Silent Valley, the moist evergreen forest from being destroyed by a hydroelectric project.



Save Silent Valley was an environmental movement intended to protect Silent Valley, an evergreen tropical forest in the Palakkad district of Kerala, India. The Kerala State Electricity Board (KSEB) proposed a hydroelectric dam across the Kunthipuzha River that runs through Silent Valley. In February 1973, the Planning Commission approved the project at a cost of about Rs 25 crores. Many feared that the project would submerge 8.3 sq km of untouched moist evergreen forest. Several NGOs strongly opposed the project and urged the government to abandon it. In January 1981, bowing to unrelenting public pressure, Indira Gandhi declared that Silent Valley will be protected. In June 1983 the Center re-examined the issue through a commission chaired by Prof. M.G.K. Menon. In November 1983 the Silent Valley Hydroelectric Project was called off. In 1985, Prime Minister Rajiv Gandhi formally inaugurated the Silent Valley National Park.

Appiko Movement

Place: Uttara Kannada and Shimoga districts of Karnataka State

Leaders: Appiko's greatest strengths lie in it being neither driven by a personality nor having been formally institutionalised. However, it does have a facilitator in Pandurang Hegde. He helped launch the movement in 1983.

Aim: Against the felling and commercialization of natural forest and the ruin of ancient livelihood.



Appiko Movement focused on the issues of forest-based environmental actions in India. Main objectives of this movement are –

- Protecting the existing forest cover.
- Regeneration of trees in denuded land.
- Utilizing forest wealth with proper consideration to conservation of natural resources.

It can be said that Appiko movement is the southern version of the Chipko movement. The Appiko Movement was locally known as “Appiko Chaluvali”. The locals embraced the trees which were to be cut by contractors of the forest department. The Appiko movement used various techniques to raise awareness such as foot marches in the interior forest, slide shows, folk dances, street plays etc. The movement became a success.

Narmada Bachao Andolan (NBA)

- Place: Narmada River, which flows through the states of Gujarat, Madhya Pradesh and Maharashtra.
- Leaders: Medha Patkar, Baba Amte, adivasis, farmers, environmentalists and human rights activists.
- Aim: A social movement against a number of large dams being built across the Narmada River.



The most widespread movement in the environmental crusade of India is the movement against the Narmada River Valley Project. . As a democratic movement, it followed the Gandhian way completely. The movement first started as a protest for not providing proper rehabilitation and resettlement for the people who have been displaced by the construction of **Sardar Sarovar Dam**. Later on, the movement turned its focus on the preservation of the environment and the eco-systems of the valley. Activists also demanded the height of the dam to be reduced to 88 m from the proposed height of 130m. World Bank withdrew from the project.

The environmental issue was taken into court. In October 2000, the Supreme Court gave a judgment approving the construction of the Sardar Sarovar Dam with a condition that height of the dam could be raised to 90 m. Although this movement was not successful as the dam could not be prevented. But this movement has created an anti-big dam opinion in India and outside. It questioned the paradigm of development.

Navdanya

Navdanya is an Indian-based non-governmental organisation which promotes biodiversity conservation, biodiversity, organic farming, the rights of farmers, and the process of seed saving.

One of Navdanya's founders, and outspoken members, is Dr. Vandana Shiva, an environmental activist, physicist, and author.

Navdanya movement is a movement for Earth Democracy based on the philosophy of '*Vasudhaiv Kutumbakam*' (The Earth as one Family). They protect the India's biodiversity based food heritage through *Bija Swaraj*, *Ann Swaraj*, *Bhu Swaraj* and *Gyan swaraj*.

Ecofeminism is the of Diverse Women for Diversity (DWD) – an international movement started in mid 90s. Navdanya or Research Foundation for science, Technology and Ecology (RFSTE) are the founding members of this movement. Diverse Women seeks to defend diversity, peace and democracy from the growing threats of monoculture, war, totalitarianism and fundamentalism.

Through '*Bija Swaraj*' Navdanya has created 122 Community Seed Banks in 18 states of India and Bhutan. They the seeds of hope to the farmers after disasters like Odisha Super Cyclone, Tsunami, Floods in Uttarakhand, Phailin in Odisha and Nepal earthquake disaster.

Through '*Anna Swaraj*' they help farmers and consumers through awareness of benefits of local, fresh organic nutritious food and training and harm to the health from Chemicals which are destroying biodiversity, trapping farmers in to the debt and suicides and spreading diseases epidemics in society

Through '*Bhu Swaraj*' they are committed to protect the living Soil - our mother Earth, our identity, the basis of our life and our livelihoods. They try to keep the soil healthy and fertile through organic and agro-ecological practices. They are also committed to keep our mother earth free from hazardous agrichemicals .

Through '*Gyan Swaraj*' they promote knowledge democracy, knowledge sovereignty and scientific research on biodiversity, agro-ecology and climate change highlighting the contribution of diverse species to our agriculture production and food security.

They spread ecological awareness and challenged the biopiracy of our traditional knowledge and indigenous biodiversity and won the cases related to Neem, Basmati and Wheat.

Air quality standards

The objectives of air quality standards are:

1. To indicate the levels of air quality necessary with an adequate margin of safety to protect the public health, vegetation and property;
2. To assist in establishing priorities for abatement and control of pollutant level;
3. To provide uniform yardstick for assessing air quality at national level;
4. To indicate the need and extent of monitoring programme.

- The 2005 World Health Organization's "WHO Air quality guidelines" offer global guidance on thresholds and limits for 4 key air pollutants that pose health risks - particulate matter (PM), ozone (O₃), nitrogen dioxide (NO₂) and sulfur dioxide (SO₂).
- The Clean Air Act in the US requires the Environment Protection Agency (EPA) to set National Ambient Air Quality Standards (NAAQS) and NAAQS has been set for six pollutants namely, carbon monoxide, lead, nitrogen dioxide, ozone, particle pollution and sulphur dioxide.
- Central Pollution Control Board (CPCB) has published the Revised National Ambient Air Quality Standards (NAAQS) in November, 2009 of 12 air pollutants namely PM₁₀, PM_{2.5}, Lead, Nickel, Arsenic, sulphur dioxide, nitrogen dioxide, ammonia, ozone, carbon monoxide, benzene and Benzo(a)Pyrene (BaP).
- The European Union has also developed air quality standards for PM_{2.5}, sulphur dioxide (SO₂), nitrogen dioxide (NO₂), PM₁₀, lead (Pb), carbon monoxide (CO), Benzene, Ozone, Arsenic (As), Cadmium (Cd), Nickel (Ni), Polycyclic Aromatic Hydrocarbons

MINAS (The Minimal National Standards)

CPCB formulates The Minimal National Standards (MINAS) specific for various categories of industries with regards to their effluent discharge (water pollutants), emissions (air pollutants), noise levels and solid waste. These standards are required to be adopted by State Governments as minimal standards.

(MINAS) for a particular industry is the effluent standard achievable by the industry by installing pollution control measures which are within the techno- economic capability of the industry.

NAAQS (National Ambient Air Quality Standards)

An air quality standard defines the maximum amount of a pollutant averaged over a specified period of time that can be present in outdoor air without harming public health, and thus, it defines clean air.

National Ambient Air Quality Standards are the standards for ambient air quality set by the Central Pollution Control Board (CPCB) that is applicable nationwide. In order to combat air pollution, it is required to identify the pollutants, its source of emission and investigate the effects of living and the environment. The Central Pollution Control Board has notified the revised National Ambient Air Quality Standards November, 2009.

The objectives of air quality standards:

- ❑ To indicate the levels of air quality necessary with an adequate margin of safety to protect the public health, vegetation and property;
- ❑ To assist in establishing priorities for abatement and control of pollutant level;
- ❑ To provide uniform yardstick for assessing air quality at national level and to indicate the need and extent of monitoring programme.

The World Health Organization (WHO) has established guidelines on outdoor (ambient) air pollution levels, which are widely used as reference tools by policymakers across the world to set standards and goals for air quality management. Initially set in 2005, they were updated in 2021. The WHO Air quality guidelines recommend levels and interim targets for common air pollutants: PM, O₃, NO₂, and SO₂.

Pollutant	Averaging Time	2005 AQGs	2021 AQGs
PM _{2.5} , µg/m ³	Annual	10	5
	24-hour ^a	25	15
PM ₁₀ , µg/m ³	Annual	20	15
	24-hour ^a	50	45
O ₃ , µg/m ³	Peak season ^b	-	60
	8-hour ^a	100	100
NO ₂ , µg/m ³	Annual	40	10
	24-hour ^a	-	25
SO ₂ , µg/m ³	24-hour ^a	20	40
CO, mg/m ³	24-hour ^a	-	4

Air Quality Index (AQI)

An air quality index is an indicator developed by government agencies to communicate to the public how polluted the air currently is or how polluted it is forecast to become. Different countries have their own air quality indices, corresponding to different national air quality standards.

As the AQI increases, it means that a large percentage of the population will experience severe adverse health effects. Children, the elderly and individuals with respiratory or cardiovascular problems are typically the first groups affected by poor air quality.

Computation of the AQI requires an air pollutant concentration over a specified averaging period, obtained from an air monitor or model. Air quality index values are typically grouped into ranges. Each range is assigned a descriptor, a color code, and a standardized public health advisory.

Environmental Impact Assessment (EIA)

UNEP defines **Environmental Impact Assessment (EIA)** as a **tool** used to **identify** the environmental, social and economic **impacts** of a **project prior to decision-making**.

According to International Association for Impact Assessment (IAIA), the definition of EIA is 'the process of identifying, predicting, evaluating and mitigating the biophysical, social and other relevant effects of development proposals prior to major decisions being taken and commitments made'.

Origin of EIA is traced to the formulation of National Environment Policy Act (NEPA) in 1969 in USA. In many countries EIA became popular with the introduction of the concept of sustainable development after the World Commission on Environment and Development in 1987 and the Rio summit in 1992. A beginning in this direction was made in our country with the impact assessment of river valley projects in 1978-79 and the scope has subsequently been enhanced to cover other developmental sectors such as industries, thermal power projects, mining schemes etc. EIA was made mandatory in India in 1994. To facilitate collection of environmental data and preparation of management plans, guidelines have been evolved and circulated to the concerned Central and State Government Departments.



Objectives of EIA :

- Promote development that is sustainable and optimize resource use and management opportunities
- Predict environmental, social, economic and cultural consequences of a proposed activity
- Ensure that environmental factors are considered in the decision-making process
- informing the public about the proposal
- ensuring that possible adverse environmental impacts are identified and avoided or minimised
- modify and improve design
- identify key impacts and measures for mitigating them
- provide provisions for the involvement of public, departments of government and other relevant agencies in the review of the proposed activities
- protect human health and safety

Principles of EIA:

- ☐ Certainty
- ☐ Transparency
- ☐ Practicability
- ☐ Reliability
- ☐ Flexibility
- ☐ Accountability
- ☐ Cost – effectiveness
- ☐ Participation



Methods in EIA

❑ **Ad hoc approach** – This is the oldest method but it lacks precision. this method provides the qualitative assessment of the total impact and suggests the broad areas of the possible impacts.

❑ **Checklist method** - checklists are environmental attributes and are generally of four types.

- (i) Simple checklist – involves potentially affecting factors
- (ii) Descriptive checklist – describes the measurement and predictive technique for each factor
- (iii) Scaling checklist – indicates the relative significance of each impact along with the details on critical values which represent a threshold of concern for each factor
- (iv) Scaling weighting checklist – in addition to scaling checklist it provides additional information for the subjective evaluation of each parameter w.r.t all other parameters.

❑ **Interaction Matrix Method** – This method uses a matrix format to relate project actions and environmental components. It identifies interactions between various project actions and environmental parameters and components. This method is flexible and serves both as a checklist and as a summary of impact assessment. This is considered as the ideal tool for rapid assessment.



❑ **Networks** – These show the interrelationships among the different aspects of the environment of the area. They also indicate flow of energy or impact through the environment. Networks can be used for showing the temporal and spatial flows of impact. Sequence diagram, directed diagrams or impact trees are examples of different types of diagrams.

❑ **Modelling** – This is a very popular impact assessment method which is very useful in the quantification of impacts in a highly precise way. It consists of a set of structurally interrelated variables that fit into a set of mathematical equations and the mathematical equations are processed with the help of computers.

❑ **Simulation modelling workshops** – In this approach, interdisciplinary team consisting of scientists and environmental managers interact with experts through modelling workshops, predict impacts and evaluate alternatives

❑ **Overlay mapping** – An overlay is based on a set of transparent maps each of which represents the spatial distribution of an environmental characteristics. These maps are overlaid to produce a composite map. The resulting composite map shows the area's social, physical, ecological, land use and other relevant characteristics relative to the position of the proposed project.

❑ **Geographic Information System**- it can be used as a tool for displaying and visualizing trends and patterns in spatial data set. GIS can be used for basic mapping of the environment or the project or particular impacts from it to provide visual aids to the planners, managers and consultants.



The EIA process

- ❑ **Screening** to determine which projects or developments require a full or partial impact assessment study. The screening process divides the project proposals into three categories :
 - (i) projects clearly requiring EIA
 - (ii) projects not requiring EIA
 - (iii) projects for which the application of EIA is not clear
- ❑ **Scoping** to identify which potential impacts are relevant to assess (based on legislative requirements, international conventions, expert knowledge and public involvement), to identify alternative solutions that avoid, mitigate or compensate adverse impacts on biodiversity (including the option of not proceeding with the development, finding alternative designs or sites which avoid the impacts, incorporating safeguards in the design of the project, or providing compensation for adverse impacts), and finally to derive terms of reference for the impact assessment.
- ❑ **Collection of baseline data:** Baseline refers to the background information on biological, physical, social and economic settings of the proposed location. Data collected on existing environmental status of the study area as well as new information from field samples. Primary data should always be supplemented with secondary data.
- ❑ **Assessment and evaluation of impacts and development of alternatives**, to predict and identify the likely environmental impacts of a proposed project or development, including the detailed elaboration of alternatives. Impacts are generally classified into two categories – quantifiable criteria and non – quantifiable criteria.



Impact analysis processes is generally done through five steps :

- ✓ Identification of potential impact
- ✓ Description of existing environment
- ✓ Procurement of relevant bylaws
- ✓ Prediction of impact
- ✓ Assessment of significant impacts

❑ **Mitigation:** This step recommends the actions to **reduce** and **avoid** the adverse environmental consequences of the proposal. This step aims to **maximize** the project **benefits** and minimize the **undesirable** impacts.

A mitigation plan should be prepared for **chosen alternative** to **reduce or avoid** the adverse environmental impacts, supplemented with **Environmental Management Plan** (EMP), **Risk Assessment Report**, **disaster** management plan, **rehabilitation plan** etc.

❑ **Public consultation:** The primary objective of this step is **public involvement**, **communication plan**, **providing information** of the project to the **stake holders** and considering their **views in decision making process**.

Components –

- i. Public hearing – to be carried out by concerned SPCB (State Pollution Control Board)/UTPCC (Union Territory Pollution Control Committee) within 45 days of request from the applicant.
- ii. Obtaining written response from the concerned people by putting the summary of EIA report and application on their website. This should be done within 7 days of the receipt of request for holding public hearing.



Interested public should get a copy of the project proposal. There should be a scope for the interested public for giving written opinions or attending public hearing.

☐ **Reporting the Environmental Impact Statement (EIS) or EIA report**

An EIA report generally contains

- (i) an executive summary of EIA findings
- (ii) a description of the project
- (iii) impacts of the project on the environment and data pertaining to how they were identified and predicted
- (iv) details of different mitigation options and the rationale behind choosing those options
- (v) an overview of gaps and uncertainties
- (vi) summary of EIA for general public

☐ **Review of the Environmental Impact Statement (EIS)**, based on the terms of reference (scoping) and public (including authority) participation.



☐ **Decision-making** on whether to approve the project or not, and under what conditions. At this stage the decision makers go through the EIA report, review of the EIA report thoroughly, consider the practical situation, put their practical experience, expertise and knowledge and finally decide on the course of action.

❑ **Monitoring, compliance, enforcement and environmental auditing.**

Monitor whether the predicted impacts and proposed mitigation measures occur as defined in the EMP. Verify the compliance of proponent with the EMP, to ensure that unpredicted impacts or failed mitigation measures are identified and addressed in a timely fashion. For every project, possible alternatives should be identified and environmental attributes compared. Alternatives should cover both project location and process technologies. Once alternatives have been reviewed, a mitigation plan should be drawn up for the selected option and is supplemented with an Environmental Management Plan (EMP) to guide the proponent towards environmental improvements. Inventory analysis and hazard probability and index also form part of EIA procedures.



Projects Requiring EIA Clearance from the Central Government

1. Industries & Manufacturing

- Petrochemical complexes
- Petroleum refineries
- Cement plants (≥ 1.0 MTPA)
- Fertilizer plants
- Pulp & paper industries ($\geq 30,000$ TPA)
- Chlor-alkali industries
- Integrated iron & steel plants
- Thermal power plants (≥ 500 MW for coal/lignite, ≥ 50 MW for other fuels)
- Nuclear power projects

• 2. Mining & Extraction

- Coal mining (≥ 50 ha)
- Mineral mining (≥ 100 ha)
- Offshore/onshore oil & gas drilling
- River valley and hydroelectric projects (≥ 50 MW)



3. Infrastructure & Transport

- Airports
- Highways & expressways
- Ports & harbors
- Railway projects
- Large urban development projects (SEZs, IT parks, etc.)
- Inland waterways

4. Waste Management & Hazardous Materials

- Hazardous waste treatment & disposal facilities
- Biomedical waste treatment plants
- Municipal solid waste management facilities

5. Large-scale Real Estate & Townships

- Building & construction projects covering $\geq 150,000$ sq. meters
- Townships & area development projects covering ≥ 50 ha

6. Water Resources & Irrigation

- Major irrigation projects ($\geq 10,000$ ha command area)
- Large dams and reservoirs



List of EIA-Exempted Projects (Under Various Notifications)

Projects with minimal potential environmental impact are usually exempted.

Small-scale industries categorized as Micro, Small, and Medium Enterprises (MSMEs), certain agricultural activities, minor infrastructure projects like small roads within villages, modernization of existing irrigation projects, small hydroelectric projects (below a certain capacity), expansion of existing industrial units within designated industrial estates, and certain mining activities with limited scale

Validity of Environmental Clearance

Validity of Prior Environmental Clearance (PEC) for

- ☐ river valley projects is for 10 years
- ☐ mining projects 30 years (maximum)
- ☐ 5 years in case of all other types of projects and activities
- ☐ the validity period of construction projects can be extended to a maximum of 5 years through timely application (along with required documents).



Benefits of EIA:

- ✓ facilitates the design of a monitoring programme
 - ✓ allows people to examine the underlying need for a project
 - ✓ increased accountability and transparency during the development process
 - ✓ gives people the opportunity to identify problems
 - ✓ helps a developer to design a more publicly acceptable and eco friendly project
 - ✓ exploration of alternatives can help identify cost-saving and other beneficial changes
- improved integration of projects into their environmental and social setting and reduces environmental damage
- ✓ provides a positive contribution toward achieving sustainability.

Drawbacks of EIA:

- ❑ Lack of reliable data source
- ❑ Lack of expertise in the team formed for conducting EIA of a particular project
- ❑ Public comments are not considered at the early stage which often leads to conflict at later stage of project clearance.
- ❑ Details regarding the effectiveness and implementation of mitigation measures are often not provided.
- ❑ The limited coverage of scoping is mainly confined to direct impacts.
- ❑ Emergency preparedness plans are not discussed in details and detailed information is not given to the communities.
- ❑ There is a lack of extensive ecological and socio – economic indicators for the assessment of impact.



Environmental Audit

An **environmental audit** is a **structured evaluation** of how an **organization's** operations, processes, and practices **affect** the environment. It ensures **compliance with laws**, **identifies risks**, and **promotes sustainable improvements**.

It is essentially an environmental management tool for measuring the effects of certain activities on the environment against set criteria or standards. It is used to

- ☐ investigate
- ☐ understand
- ☐ identify

The key concepts of environmental audit are:

Verification: audits evaluate compliance to regulations or other set criteria.

Systematic: audits are carried out in a planned and methodical manner.

Periodic: audits are conducted to an established schedule.

Objective: information gained from the audit is reported free of opinions.

Documented: notes are taken during the audit and the findings recorded.

Management tool: audits can be integrated into the management system (such as a quality management system or environmental management system).



The International Chamber of Commerce (ICC, 1989) defines environmental auditing as: A management tool comprising systematic, documented, periodic and objective evaluation of how well environmental organisation, management and equipment are performing with the aim of helping to safeguard the environment by facilitating management control of practices and assessing compliance with company policies, which would include regulatory requirements and standards applicable.

The key objectives of an environmental audit are to:

- Determine how well the environmental management information systems and equipment are performing
- Create the efficient and effective control mechanisms over the implementation of the national environmental legislation and environmental policy
- Verify compliance with the relevant national, local or other laws and regulations
- Minimize human exposure to risks from environmental, health and safety problems

Types of environmental audits –

- **Environmental Management Audit** - specifically designed to check and evaluate the effectiveness of environmental management systems
- **Environmental Compliance Audit** – specifically designed to test compliance (which covers both legal compliance and corporate compliance) to environmental policies, objectives, laws, by-laws, ordinances, regulations and standards.
- **Environmental Assessment Audit** - is an instrument used to check that an Environmental Impact Assessment complies with the minimum legal requirements and also checks to ensure that due legal process has been followed.
- **Waste Audit** – specifically look at the waste management component of an operation or site. It reviews various aspects of waste management and verify the methods, procedures and systems.
- **Environmental Due Diligence Audit** - looks at the actual and potential environmental liabilities of a site or operation.
- **Supplier Audit** – is carried out by a client to test the environmental compliance of a contractor or supplier.

ISO standards

ISO 14000, which was initially released in 1996, is a global series of environmental management systems (EMS) standards.

ISO 14000 history

- ✓ The first environmental management system standard, BS 7750, was published in 1992 by the BSI group
- ✓ In 1996, the International Organization for Standardization (ISO) created the ISO 14000 family of standards
- ✓ ISO 14001 underwent revision in 2004
- ✓ The current revision of ISO 14001 was published in September 2015

The ISO 14000 series addresses the following aspects of environmental management:

- ☐ Environmental Management Systems (EMS)
- ☐ Environmental Auditing & Related Investigations (EA&RI)
- ☐ Environmental Labels and Declarations (EL)
- ☐ Environmental Performance Evaluation (EPE)
- ☐ Life Cycle Assessment (LCA)
- ☐ Terms and Definitions (T&D)

An EMS is defined by ISO as: "part of the overall management system, that includes organizational structure, planning activities, responsibilities, practices, procedures, processes, and resources for developing, implementing, achieving, and maintaining the environmental policy."

Objectives of ISO 14000 series-

- ☐ To result in better environmental management
- ☐ To encompass environmental management systems and the environmental aspects of products
- ☐ To be applicable in all countries
- ☐ To promote the broader interests of the public as well as users of these standards
- ☐ To be cost-effective, non-prescriptive and flexible so they are able to meet the differing needs of organizations of any type or size, worldwide
- ☐ As part of their flexibility, to be suitable for internal and/or external verification
- ☐ To be scientifically based
- ☐ To be practical, useful and usable

The ISO 14000 family is a comprehensive set of standards that guide organizations in managing their environmental responsibilities.

Key Standards in the ISO 14000 Family:

Standard	Focus area
ISO 14001	Environmental Management Systems (EMS) – Requirements and framework for implementation.
ISO 14004	General guidelines on EMS principles, systems, and support techniques.
ISO 14005	Phased approach to EMS implementation and performance evaluation.
ISO 14006	Integration of eco-design into EMS.
ISO 14015	Environmental assessment of sites and organizations (EASO)
ISO 14020–14025	Environmental labeling and declarations (Type I, II, III).
ISO 14031	Environmental performance evaluation – measurement and reporting.
ISO 14040–14044	Life Cycle Assessment (LCA) principles, framework, and requirements.
ISO 14046	Water footprint assessment.
ISO 14064–14067	Greenhouse gas (GHG) quantification, reporting, verification, and carbon footprint.
ISO 19011	Guidelines for auditing management systems (including EMS).

ISO 14001:

- Defines requirements for an Environmental Management System (EMS).
- Provides a framework for organizations to identify, control, and improve environmental impact.
- Certification is possible through accredited third-party auditors.
- Focuses on continuous improvement, requiring organizations to identify, manage, and reduce their environmental footprint rather than just meeting minimum performance levels.
- covers environmental policy, planning (identifying environmental impacts), implementation, measurement/evaluation, and management review.
- Designed for any type of organization, regardless of size or sector.

Benefits of ISO 14001 are

- Improve resource efficiency
- Reduce waste and drive down costs
- Provide assurance that environmental impact is being measured
- Gain competitive advantage in supply chain design
- Increase new business opportunities
- Meet legal obligations
- Increase stakeholder and customer trust
- Improve overall environmental impact
- Manage environmental obligations with consistency

ISO 14001 has become one of the most successful and widely adopted standards in the ISO 14000 family because it delivers practical, measurable benefits to organizations while aligning with global priorities on sustainability.

- **Universal Applicability** - ISO 14001 is designed to be flexible and works for organizations of all sizes, sectors, and geographies. Whether it's a small business or a multinational corporation, the framework can be scaled to fit.
- **Tangible Benefits:** Organizations often see reduced waste, improved energy efficiency, and lower operational costs.
- **Operational Efficiency & Financial Gains**
By streamlining resource use, cutting down on waste, and lowering energy consumption, organizations often achieve notable reductions in operating costs. These economic benefits have been a major driver of widespread adoption.
- **Market Advantage:** Certification helps companies meet supplier requirements, enhance their reputation, and build trust with stakeholders, customers, and regulators.
- **Widespread Adoption:** Since its inception in 1996, it has been adopted by hundreds of thousands of organizations globally to manage environmental impact. As an internationally recognized standard, ISO 14001 provides consistency across borders, which is especially valuable for multinational companies.
- **Continuous Improvement framework:** The standard forces a "Plan-Do-Check-Act" (PDCA) cycle that ensures continual improvement of environmental performance, not just a one-time change.

Earth Hour

Earth Hour is a worldwide movement organized by the World Wide Fund for Nature (WWF). The event is held annually encouraging individuals, communities, and businesses to turn off non-essential electric lights for one hour, from 8:30 to 9:30 pm on a specific day towards the end of March, as a symbol of commitment to the planet.

The 2007 Earth Hour was held on March 31 in Sydney, Australia at 7:30 pm, local time.

In October 2007 San Francisco ran its own "Lights Out" program inspired by the Sydney Earth Hour.

Earth Hour 2008 was held internationally on March 29, 2008 from 8 pm to 9 pm local time, marking the first anniversary of the event. 35 countries around the world participated as official flagship cities and over 400 cities also supported.

Earth Hour 2026

Date: March 28, 2026

Time: 8:30 p.m. – 9:30 p.m. (local time)

Theme: The movement encourages everyone to "Give an Hour for Earth" by participating in eco-friendly activities, such as enjoying nature, cooking sustainable meals, or, in 2026, celebrating two decades of collective action.



Carbon sequestration

The term “carbon sequestration” is used to describe both natural and deliberate processes by which CO₂ is either removed from the atmosphere or diverted from emission sources and stored in the ocean, terrestrial environments (vegetation, soils, and sediments), and geologic formations.

- ❑ **Oceanic carbon sequestration-** The world’s oceans are the primary long-term sink for human-caused CO₂ emissions. It naturally through chemical reactions between seawater and CO₂ in the atmosphere but makes the ocean water more acidic.
- ❑ **Terrestrial carbon sequestration** - sometimes termed “biological sequestration” is typically accomplished through forest and soil conservation practices that enhance the storage of carbon (such as restoring and establishing new forests, wetlands, and grasslands) or reduce CO₂ emissions (such as reducing agricultural tillage and suppressing wildfires)
- ❑ **Geologic carbon sequestration-** Geologic sequestration begins with capturing CO₂ from the exhaust of fossil fuel power plants and other major sources. The captured CO₂ is piped 1 to 4 kilometer below the land surface and injected into porous rock formations.

Green Buildings

Green building is a resource-efficient method of construction that produces healthier buildings which have less impact on the environment and cost less to maintain. It refers to the design, construction, and operation of buildings in an environmentally responsible and resource-efficient manner. The objective is to minimize both the environmental footprint and any adverse effects on human health across the entire lifespan of the building by

- Utilization of environmentally friendly materials such as recycled, renewable, or locally sourced resources to minimize environmental impact during construction.
- Incorporation of energy-saving technologies and design strategies to minimize energy consumption and reduce greenhouse gas emissions.
- Minimization water usage and promote water conservation.
- Implementation of waste management strategies to minimize landfill waste and promote resource efficiency
- Incorporation of ventilation systems, low-emission materials, and adequate daylighting to enhance indoor air quality and occupant comfort
- Integration of on-site renewable energy sources to reduce reliance on fossil fuels and mitigate carbon emissions
- Life Cycle Assessment of the building materials and system
- Adoption of green building certification (like LEED etc) systems to ensure adherence to sustainable building standards and performance criteria.

Environmental Monitoring and Management

Remote Sensing
Air and Water Quality Monitoring
Predictive modeling
Energy Optimization
Waste Management

Environmental and Health applications of IT (Information Technology) and AI (Artificial Intelligence)

Disaster Response and Management

Early Warning Systems
Disaster Response and Management
Resilience Planning

Public Health and Healthcare

Disease Surveillance
Diagnosis and Treatment
Healthcare Management
Telemedicine

Applications of AI in healthcare –

- Used for imaging medical images for diagnosis of various diseases like cancers, fractures, and neurological disorders.
- Used for predicting the likelihood of different diseases
- It helps to design personalized treatment plans by analyzing a patient's genetics, lifestyle, and medical history and can monitor patient responses to treatments and adjust medications or therapies in real-time.
- AI-powered robotic systems assist in surgeries, improving precision and reducing human error and detects early signs of depression, anxiety, and schizophrenia
- tracks and predicts disease outbreaks by analyzing global health data.
- streamlines administrative tasks, reducing the workload for healthcare professionals



Applications of AI in Environmental Conservations

- ❑ AI models analyze climate data to predict temperature changes, extreme weather events, and sea level rise.
- ❑ AI-powered sensors detect pollutants in real time and predict air quality trends and thus help in controlling pollution
- ❑ AI-driven camera traps and drones help monitor endangered species and poaching activities and Wildlife Conservation & Biodiversity Protection
- ❑ Helps in renewable energy optimization
- ❑ Helps in deforestation & Land Use Monitoring by analyzing satellite images to track forest cover loss and illegal deforestation.
- ❑ Helps disaster prediction & management and improves disaster response.
- ❑ AI helps in sustainable agriculture by optimizes irrigation, fertilizer use, and pest control thereby reducing environmental impact.
- ❑ Monitors ocean health using drones and underwater robots for tracking track marine pollution and coral reef health.